Pikachu Metrics 数据分析 metrics_Mjlab-Tracking-Flat-Pikachu-27dof-Quad-Walk__model_4999_20260911_004427_with_target

FILE
metrics_Mjlab-Tracking-Flat-Pikachu-27dof-Quad-Walk__model_4999_20260911_004427_with_target.csv

NETWORK
Mjlab-Tracking-Flat-Pikachu-27dof-Quad-Walk__model_4999

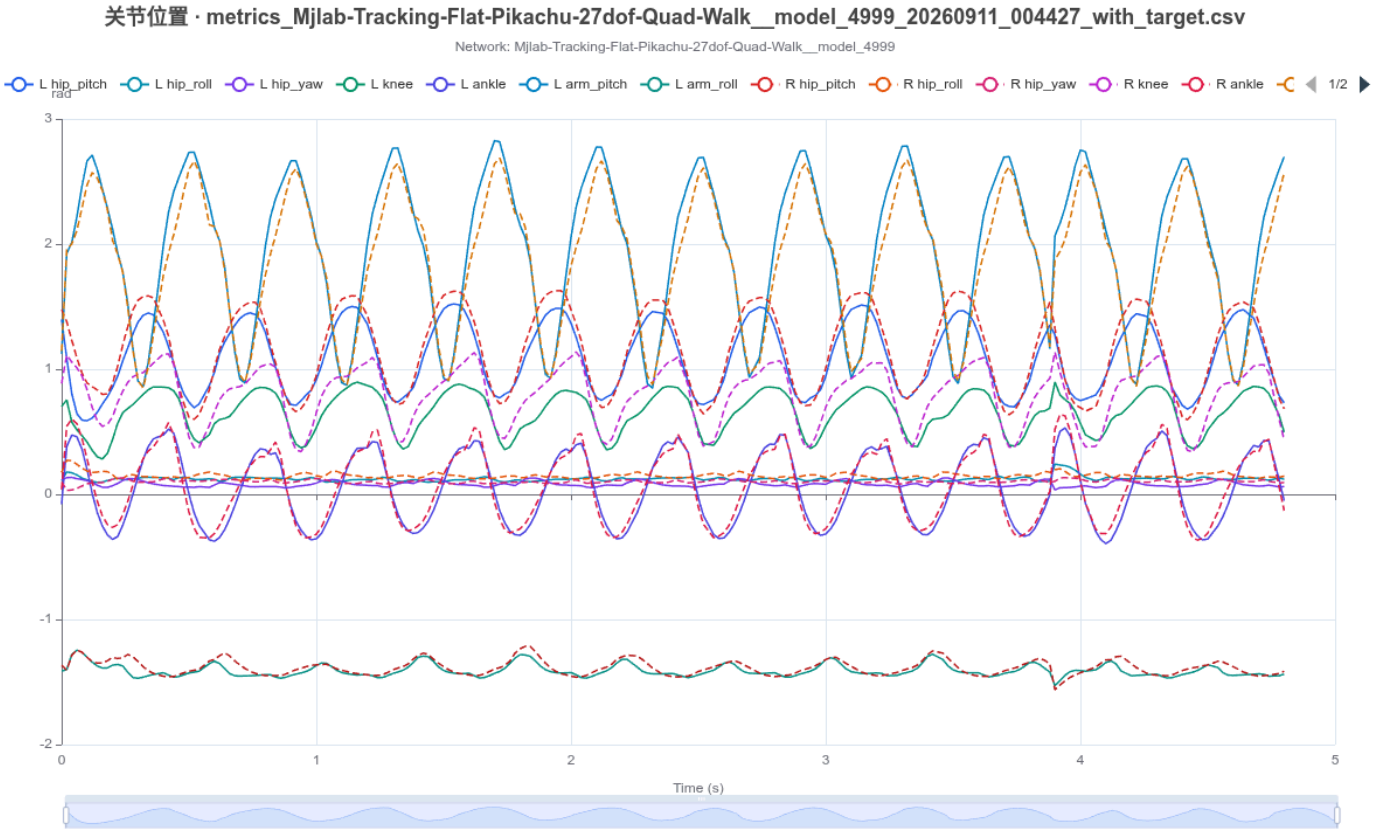
SAMPLES
221

DURATION
4.800 s

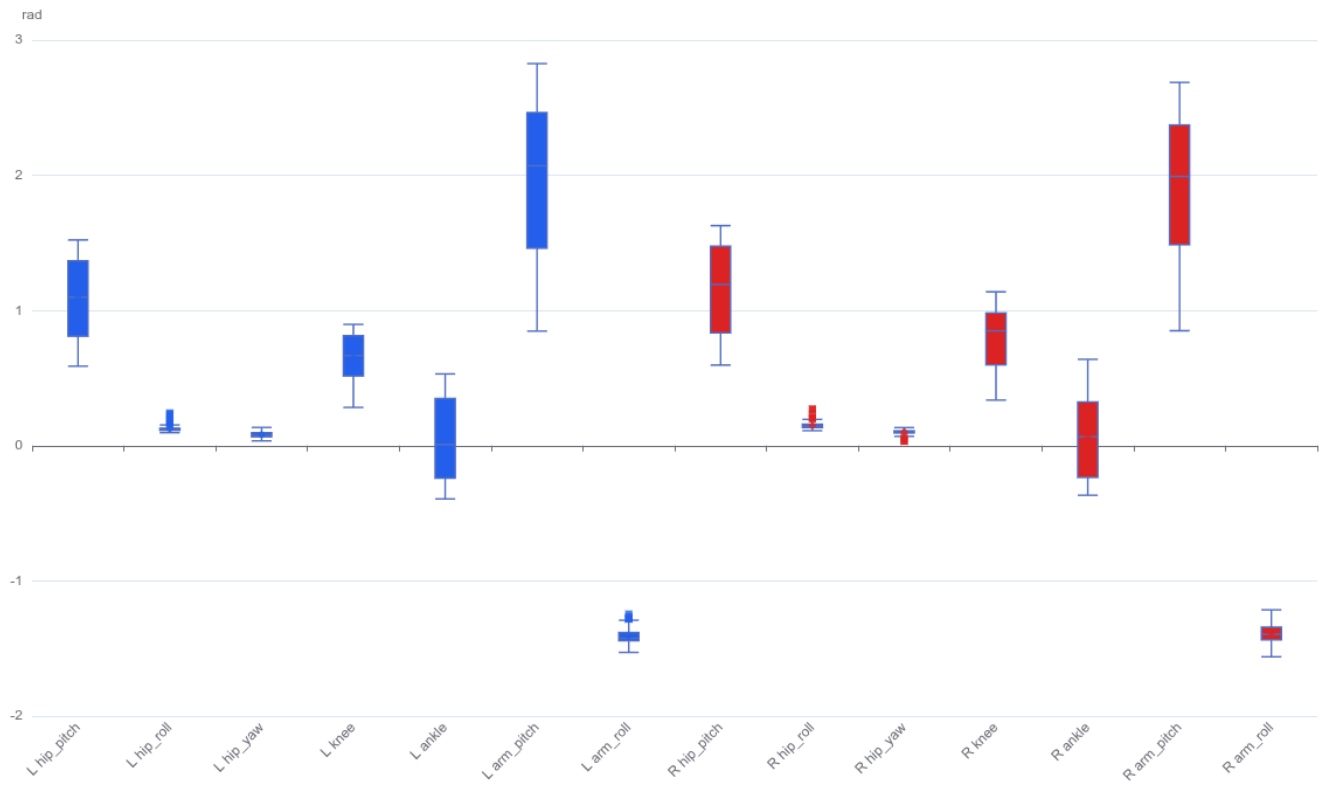
ENVIRONMENTS
1

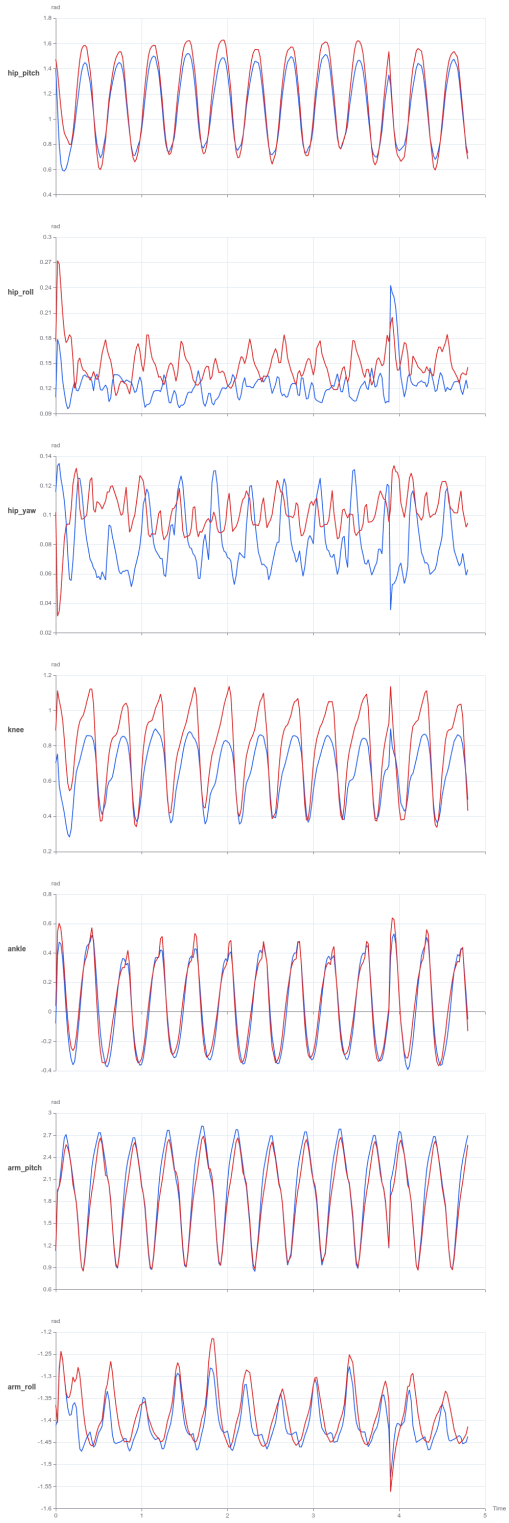
时间区间: 0.000 — 4.800 s · 仅含已勾选字段

关节位置

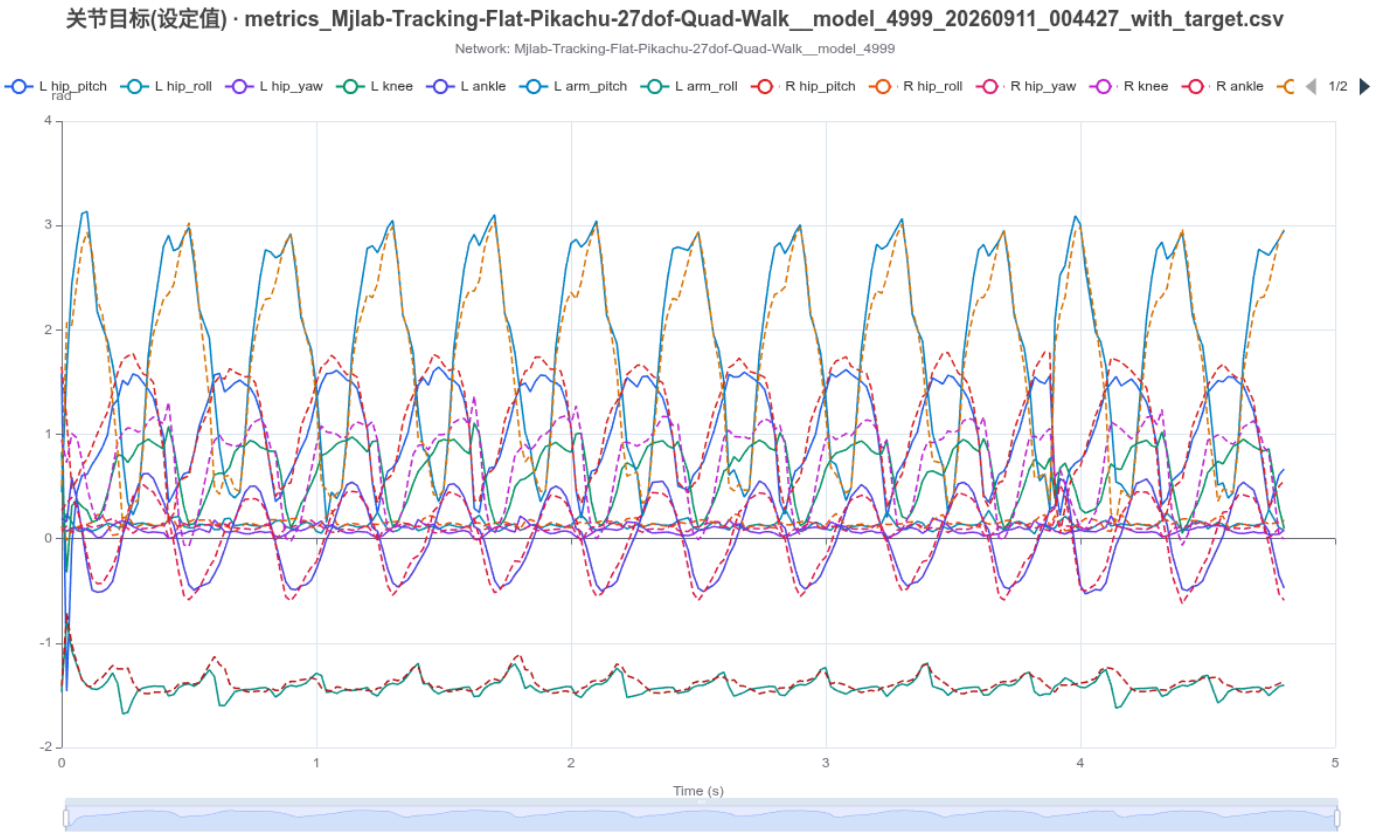


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_position_rad/left_hip_pitch_joint	30	10	221	0.588426	1.52179	1.09103	1.12597	1.52179	1.48513
joint_position_rad/left_hip_roll_joint	10	15	221	0.0958494	0.242827	0.123693	0.125249	0.242827	0.141451
joint_position_rad/left_hip_yaw_joint	10	15	221	0.0354603	0.135206	0.0825258	0.0854019	0.135206	0.124776
joint_position_rad/left_knee_joint	30	10	221	0.282781	0.897786	0.651858	0.673137	0.897786	0.859362
joint_position_rad/left_ankle_joint	10	15	221	-0.393508	0.530811	0.0449636	0.293684	0.530811	0.459096
joint_position_rad/left_arm_pitch_joint	10	15	221	0.847646	2.82652	1.97885	2.06632	2.82652	2.73733
joint_position_rad/left_arm_roll_joint	10	15	221	-1.52869	-1.24475	-1.40837	1.40923	1.52869	1.46441
joint_position_rad/right_hip_pitch_joint	30	10	221	0.596151	1.62772	1.15429	1.20026	1.62772	1.59136
joint_position_rad/right_hip_roll_joint	10	15	221	0.111414	0.272236	0.149062	0.150731	0.272236	0.183780
joint_position_rad/right_hip_yaw_joint	10	15	221	0.0312886	0.133743	0.101950	0.102920	0.133743	0.124725
joint_position_rad/right_knee_joint	30	10	221	0.337295	1.13788	0.790306	0.823734	1.13788	1.09365
joint_position_rad/right_ankle_joint	10	15	221	-0.366649	0.638968	0.0623664	0.294888	0.638968	0.511395
joint_position_rad/right_arm_pitch_joint	10	15	221	0.850282	2.68716	1.88606	1.96209	2.68716	2.60975
joint_position_rad/right_arm_roll_joint	10	15	221	-1.56193	-1.21441	-1.38449	1.38582	1.56193	1.45486

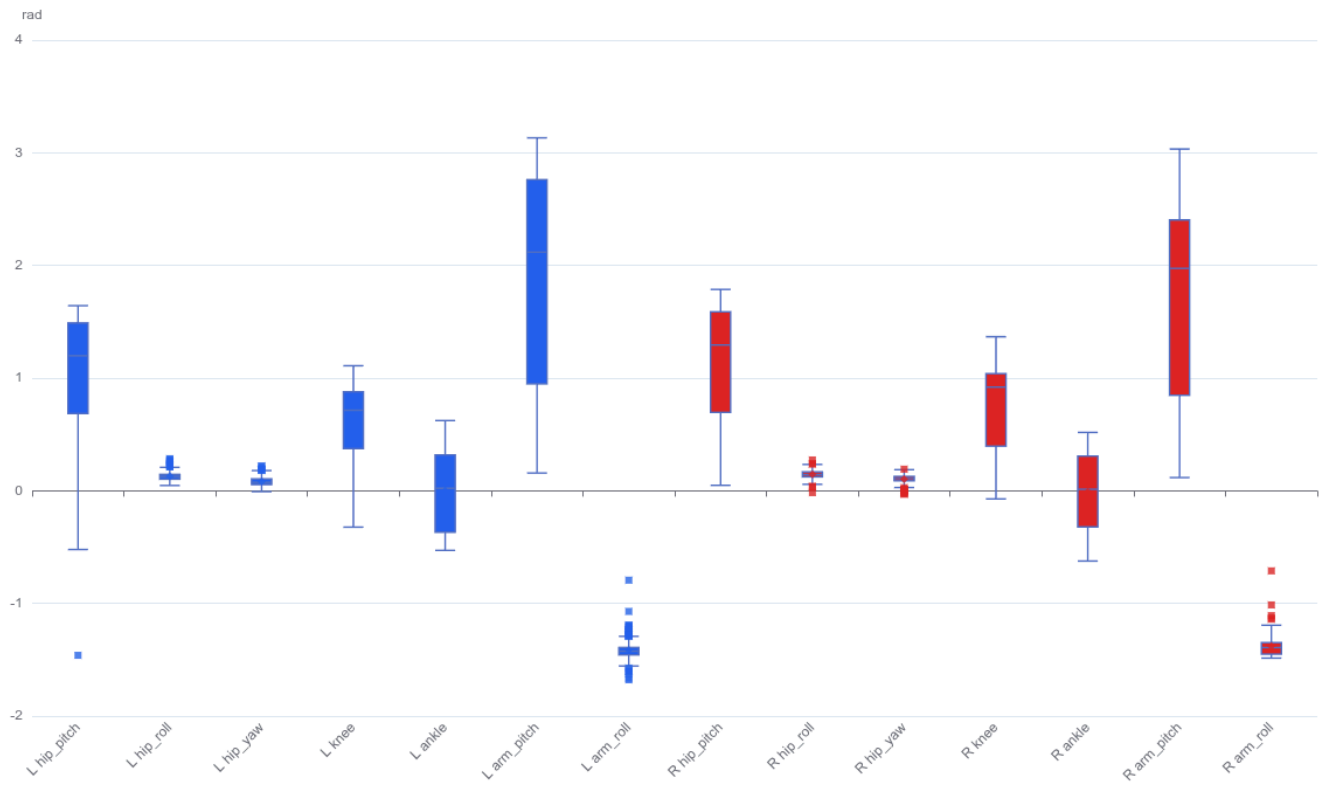

rad
3
2
1
0
-1
-2
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

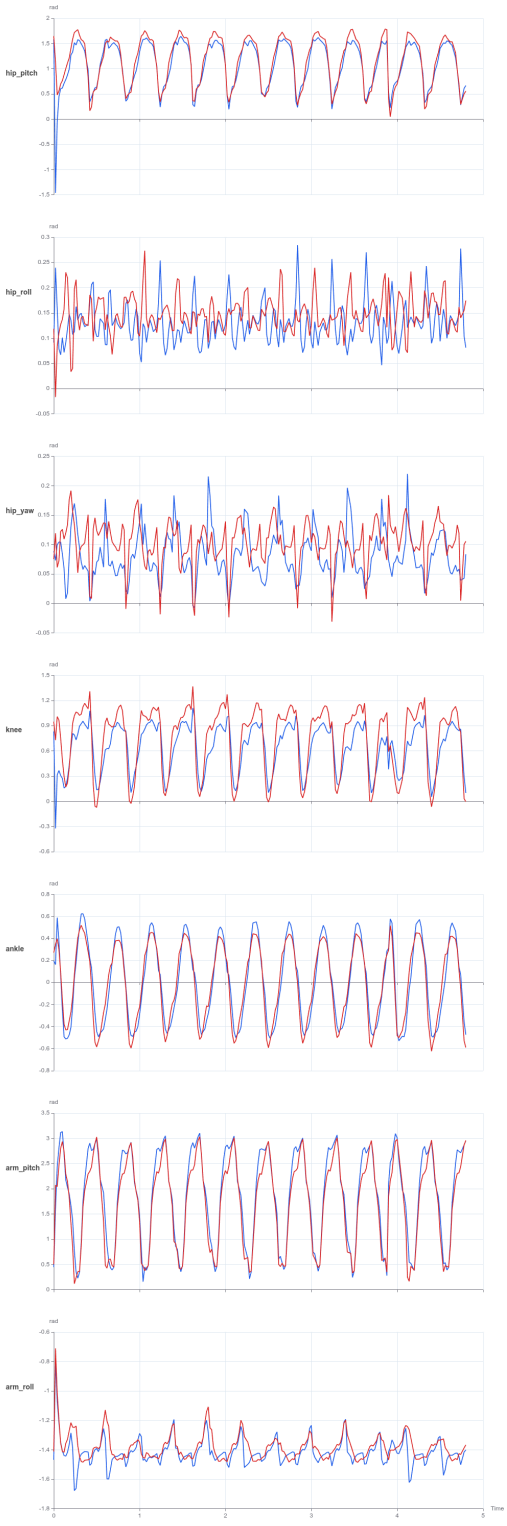

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad
Time

关节目标(设定值)

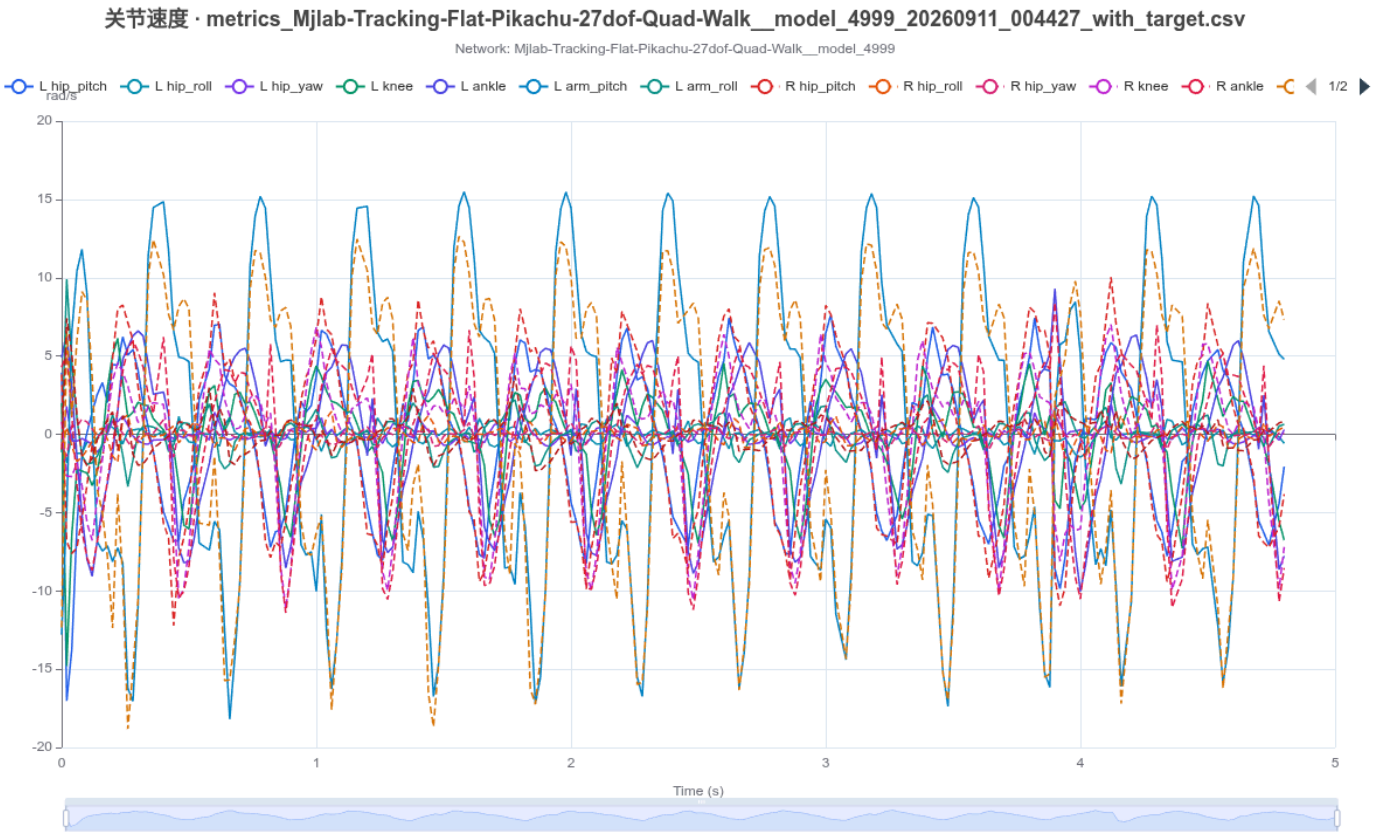


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_target_rad/left_hip_pitch_joint	30	10	221	-1.46075	1.64331	1.07508	1.16932	1.64331	1.57768
joint_target_rad/left_hip_roll_joint	10	15	221	0.0464549	0.284535	0.129596	0.135807	0.284535	0.206226
joint_target_rad/left_hip_yaw_joint	10	15	221	-0.00741272	0.220058	0.0816250	0.0914873	0.220058	0.160180
joint_target_rad/left_knee_joint	30	10	221	-0.323000	1.10925	0.635176	0.696224	1.10925	0.950798
joint_target_rad/left_ankle_joint	10	15	221	-0.529267	0.622567	-0.000269978	0.361085	0.622567	0.539221
joint_target_rad/left_arm_pitch_joint	10	15	221	0.157263	3.13291	1.88913	2.10223	3.13291	2.98051
joint_target_rad/left_arm_roll_joint	10	15	221	-1.67797	-0.793778	-1.41442	1.41740	1.67797	1.52054
joint_target_rad/right_hip_pitch_joint	30	10	221	0.0486105	1.78726	1.15926	1.25413	1.78726	1.73892
joint_target_rad/right_hip_roll_joint	10	15	221	-0.0167386	0.273394	0.146673	0.151411	0.273394	0.217067
joint_target_rad/right_hip_yaw_joint	10	15	221	-0.0311887	0.191568	0.103197	0.110154	0.191568	0.153008
joint_target_rad/right_knee_joint	30	10	221	-0.0728292	1.36723	0.744314	0.840841	1.36723	1.16027
joint_target_rad/right_ankle_joint	10	15	221	-0.623655	0.517483	-0.00928150	0.344686	0.623655	0.542006
joint_target_rad/right_arm_pitch_joint	10	15	221	0.117032	3.03324	1.73961	1.95149	3.03324	2.94263
joint_target_rad/right_arm_roll_joint	10	15	221	-1.48538	-0.711008	-1.37983	1.38299	1.48538	1.47304

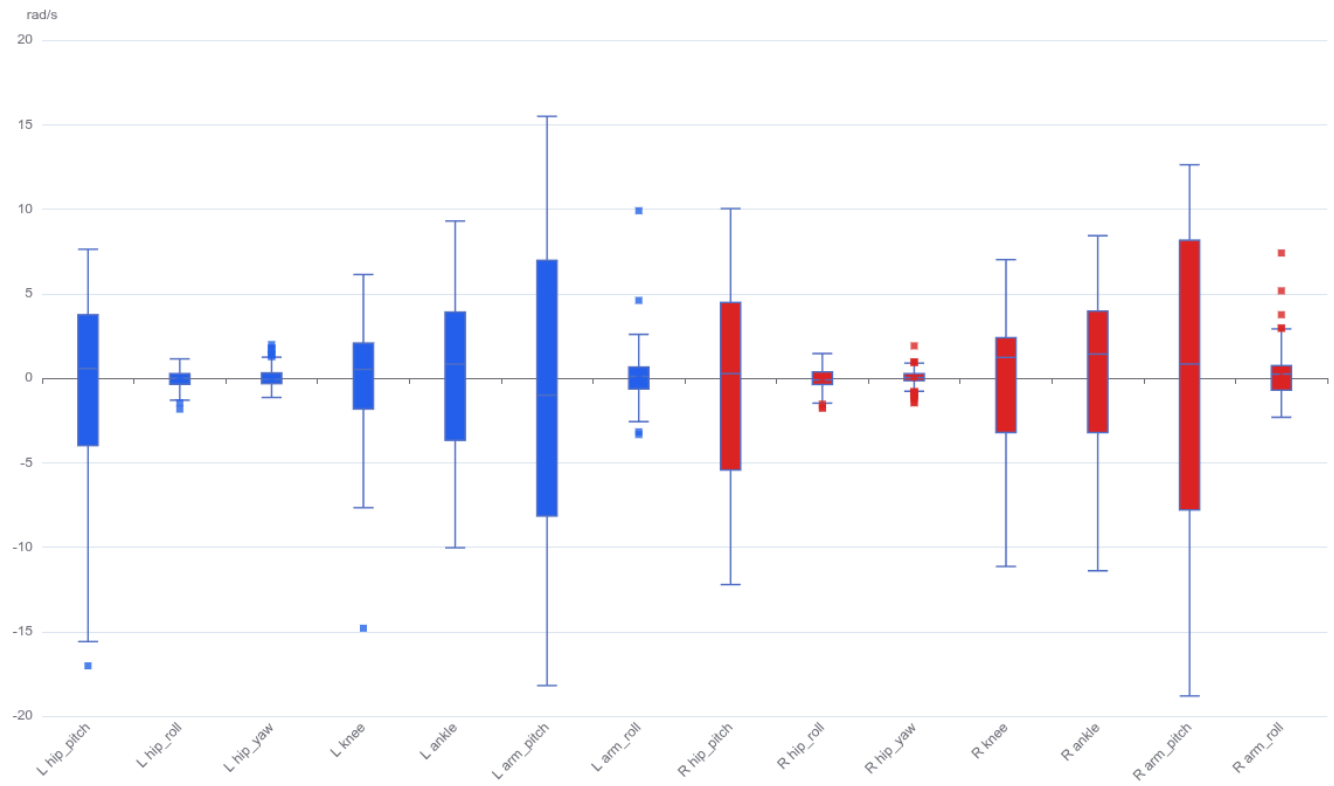

rad
4
3
2
1
0
-1
-2
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

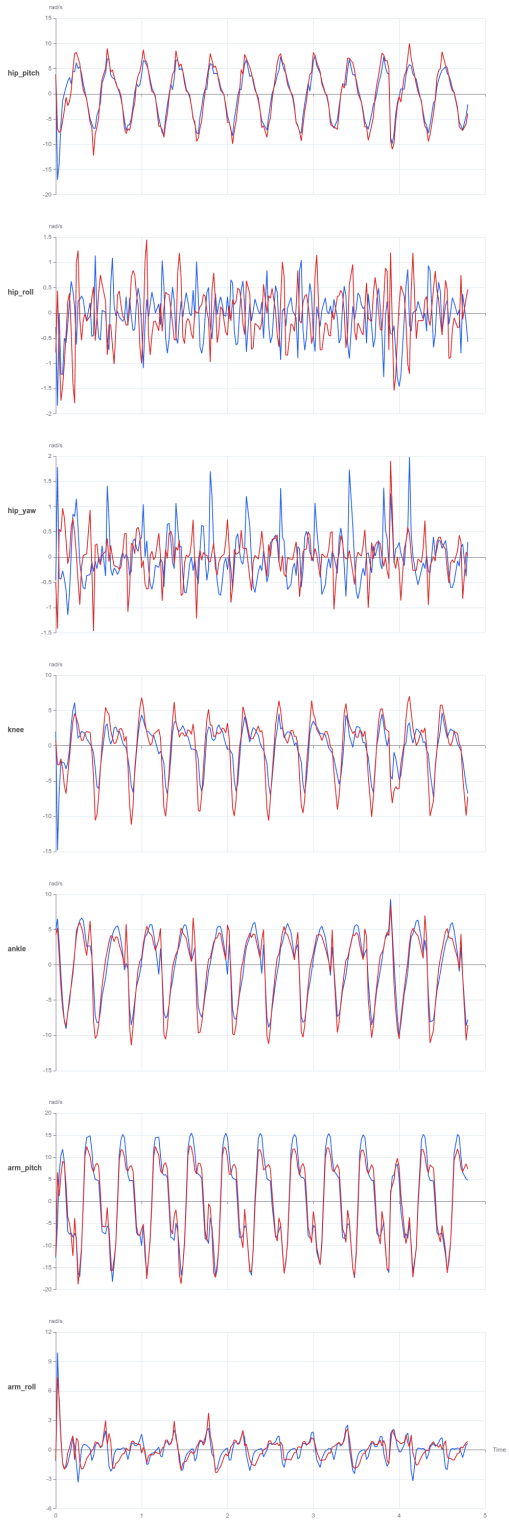

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad
Time

关节速度



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_rad_s/left_hip_pitch_joint	30	10	221	-17.0249	7.62546	-0.124927	4.59045	17.0249	7.68646
joint_velocity_rad_s/left_hip_roll_joint	10	15	221	-1.83767	1.13812	-0.0514421	0.490260	1.83767	1.03850
joint_velocity_rad_s/left_hip_yaw_joint	10	15	221	-1.14804	1.98480	0.0174293	0.516318	1.98480	1.14804
joint_velocity_rad_s/left_knee_joint	30	10	221	-14.8012	6.12735	-0.154802	3.09654	14.8012	6.12735
joint_velocity_rad_s/left_ankle_joint	10	15	221	-10.0357	9.29793	-0.120441	4.60937	10.0357	8.14398
joint_velocity_rad_s/left_arm_pitch_joint	10	15	221	-18.1956	15.4886	-0.208846	9.88715	18.1956	15.9201
joint_velocity_rad_s/left_arm_roll_joint	10	15	221	-3.34036	9.90544	0.0429177	1.28024	9.90544	2.12526
joint_velocity_rad_s/right_hip_pitch_joint	30	10	221	-12.2170	10.0322	-0.121265	5.34249	12.2170	8.79758
joint_velocity_rad_s/right_hip_roll_joint	10	15	221	-1.78821	1.45332	-0.0328340	0.568193	1.78821	1.18862
joint_velocity_rad_s/right_hip_yaw_joint	10	15	221	-1.46756	1.90692	0.0165096	0.424534	1.90692	0.919741
joint_velocity_rad_s/right_knee_joint	30	10	221	-11.1481	7.01366	-0.193785	4.36836	11.1481	9.11360
joint_velocity_rad_s/right_ankle_joint	10	15	221	-11.4009	8.42480	-0.218561	5.04814	11.4009	9.91340
joint_velocity_rad_s/right_arm_pitch_joint	10	15	221	-18.8093	12.6282	-0.185155	9.21486	18.8093	15.8277
joint_velocity_rad_s/right_arm_roll_joint	10	15	221	-2.31841	7.40443	0.0786088	1.19470	7.40443	1.98828

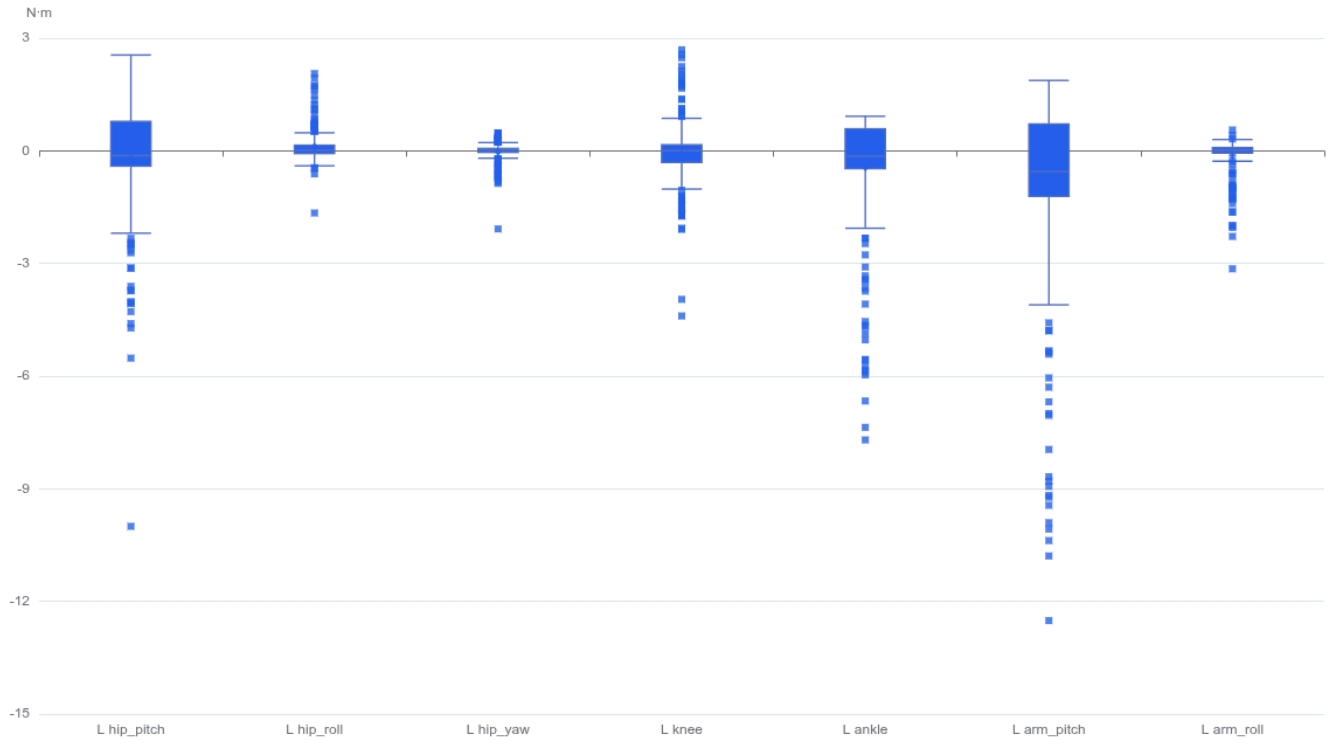

rad/s
20
15
10
5
0
-5
-10
-15
-20
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

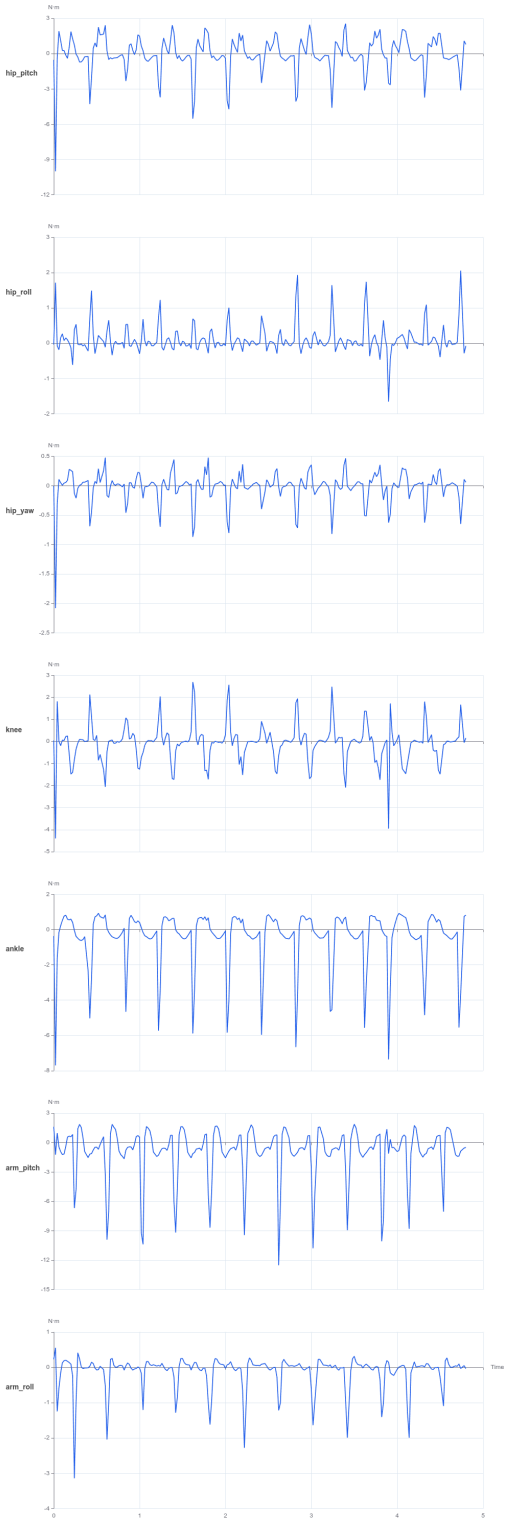

rad/s
hip_pitch
rad/s
hip_roll
rad/s
hip_yaw
rad/s
knee
rad/s
ankle
rad/s
arm_pitch
rad/s
arm_roll
Time

关节扭矩

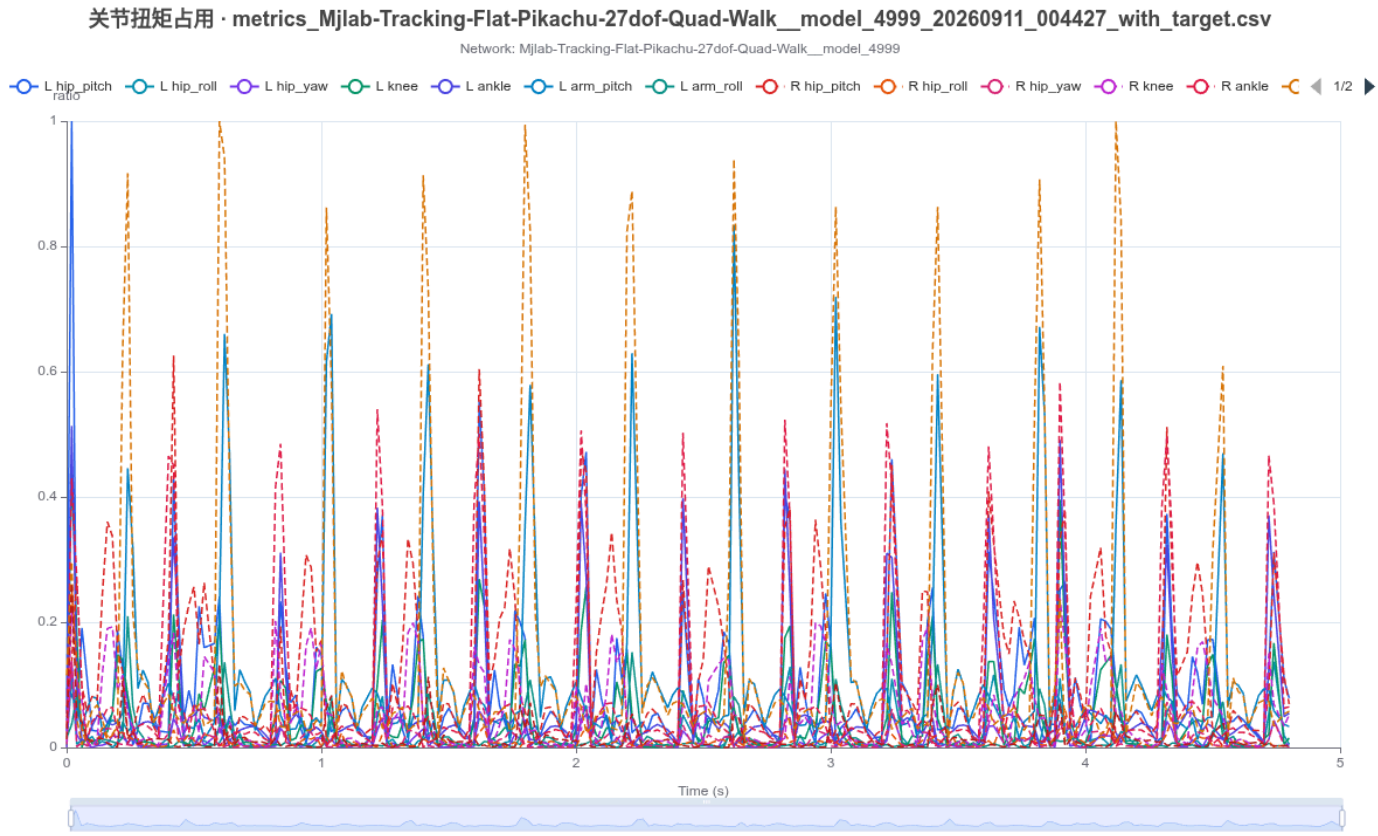


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_nm/left_hip_pitch_joint	30	10	221	-10.0000	2.54908	-0.0550562	1.52700	10.0000	3.12426
joint_torque_nm/left_hip_roll_joint	10	15	221	-1.65625	2.05615	0.108800	0.425783	2.05615	1.00649
joint_torque_nm/left_hip_yaw_joint	10	15	221	-2.08119	0.476740	-0.0251106	0.271070	2.08119	0.627276
joint_torque_nm/left_knee_joint	30	10	221	-4.39903	2.68890	-0.0829646	0.889420	4.39903	1.86212
joint_torque_nm/left_ankle_joint	10	15	221	-7.69997	0.920007	-0.437407	1.69901	7.69997	4.66407
joint_torque_nm/left_arm_pitch_joint	10	15	221	-12.5111	1.87244	-0.927393	2.80380	12.5111	7.95327
joint_torque_nm/left_arm_roll_joint	10	15	221	-3.14162	0.553887	-0.122225	0.505231	3.14162	1.21602

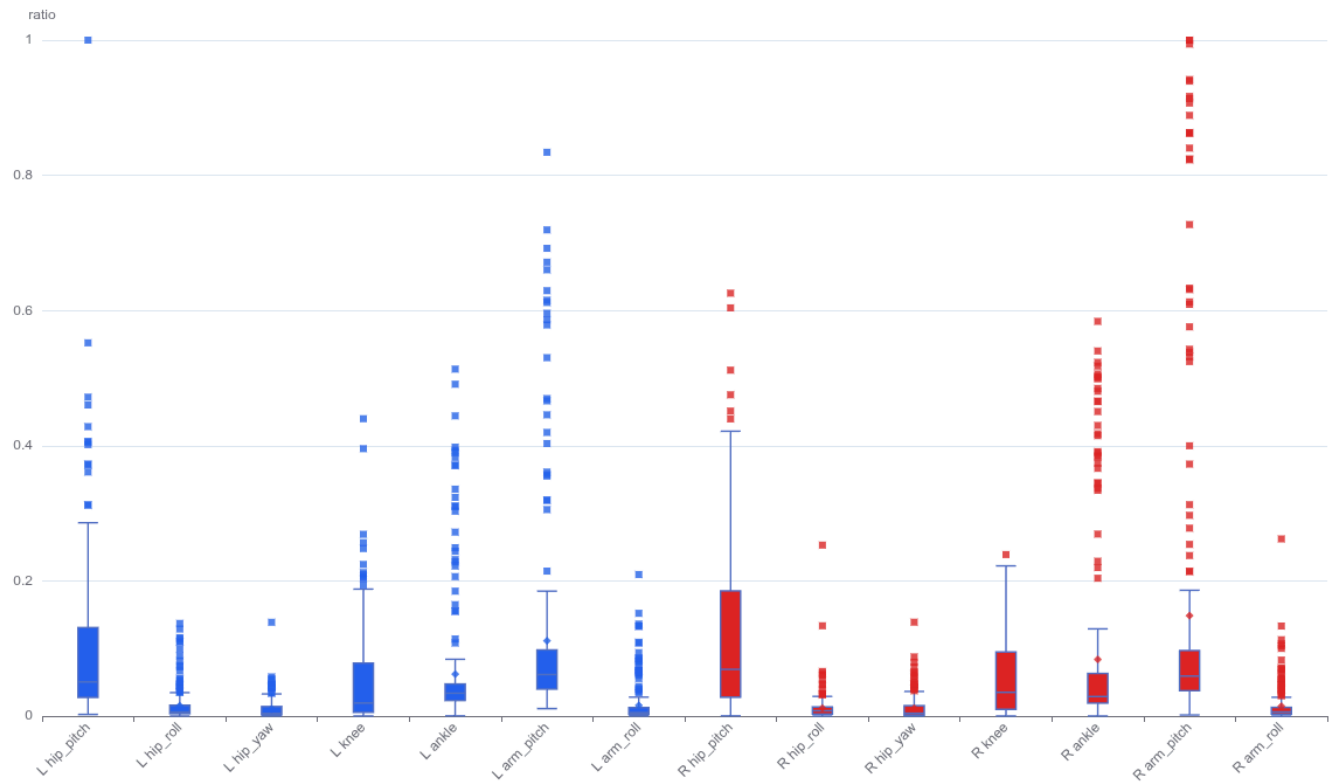

N·m
3
0
-3
-6
-9
-12
-15
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll

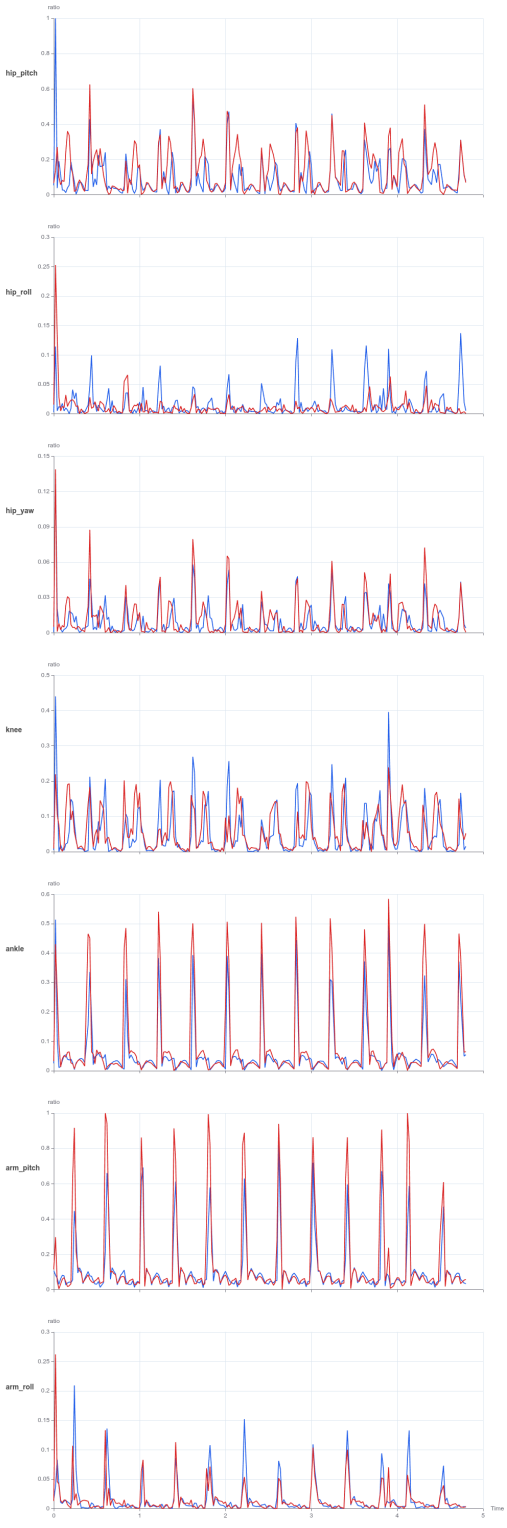

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
N·m
Time

关节扭矩占用

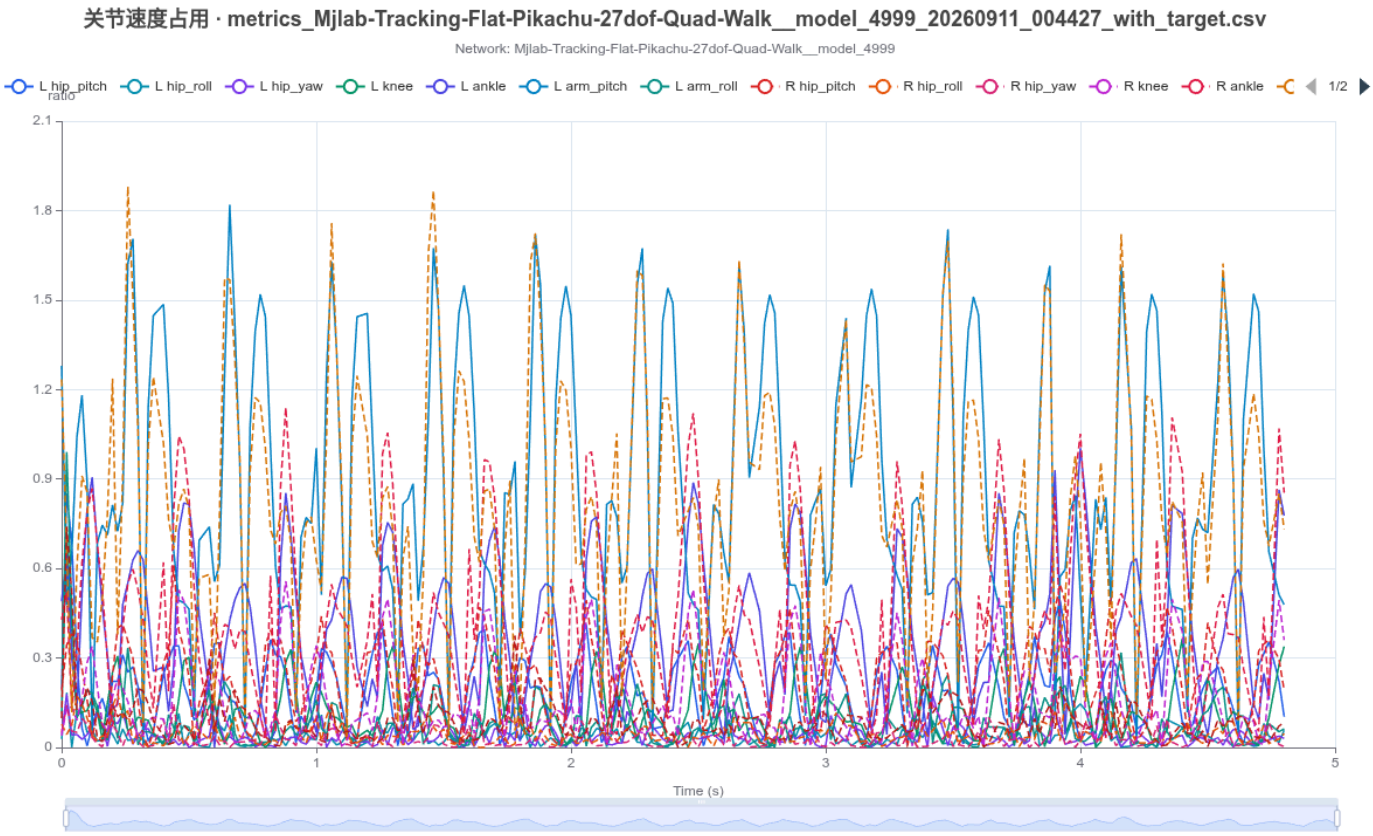


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_ratio/left_hip_pitch_joint	30	10	221	0.00239373	1.00000	0.0975643	0.152700	1.00000	0.312426
joint_torque_ratio/left_hip_roll_joint	10	15	221	0.000142955	0.137077	0.0153695	0.0283855	0.137077	0.0670993
joint_torque_ratio/left_hip_yaw_joint	10	15	221	0.0000735753	0.138746	0.0101642	0.0180714	0.138746	0.0418184
joint_torque_ratio/left_knee_joint	30	10	221	0.0000341392	0.439903	0.0526951	0.0889420	0.439903	0.186212
joint_torque_ratio/left_ankle_joint	10	15	221	0.000266325	0.513332	0.0620758	0.113267	0.513332	0.310938
joint_torque_ratio/left_arm_pitch_joint	10	15	221	0.0110072	0.834073	0.111255	0.186920	0.834073	0.530218
joint_torque_ratio/left_arm_roll_joint	10	15	221	0.0000599790	0.209442	0.0157917	0.0336821	0.209442	0.0810679
joint_torque_ratio/right_hip_pitch_joint	30	10	221	0.000375004	0.625663	0.121289	0.173338	0.625663	0.360401
joint_torque_ratio/right_hip_roll_joint	10	15	221	0.0000257586	0.252937	0.0117372	0.0242621	0.252937	0.0319384
joint_torque_ratio/right_hip_yaw_joint	10	15	221	0.0000195949	0.138889	0.0115910	0.0211750	0.138889	0.0468767
joint_torque_ratio/right_knee_joint	30	10	221	0.000137147	0.238933	0.0585668	0.0840530	0.238933	0.183038
joint_torque_ratio/right_ankle_joint	10	15	221	0.000163378	0.583925	0.0839026	0.159935	0.583925	0.465214
joint_torque_ratio/right_arm_pitch_joint	10	15	221	0.00170426	1.00000	0.148503	0.280747	1.00000	0.861898
joint_torque_ratio/right_arm_roll_joint	10	15	221	0.00000929804	0.262079	0.0145853	0.0306649	0.262079	0.0627029

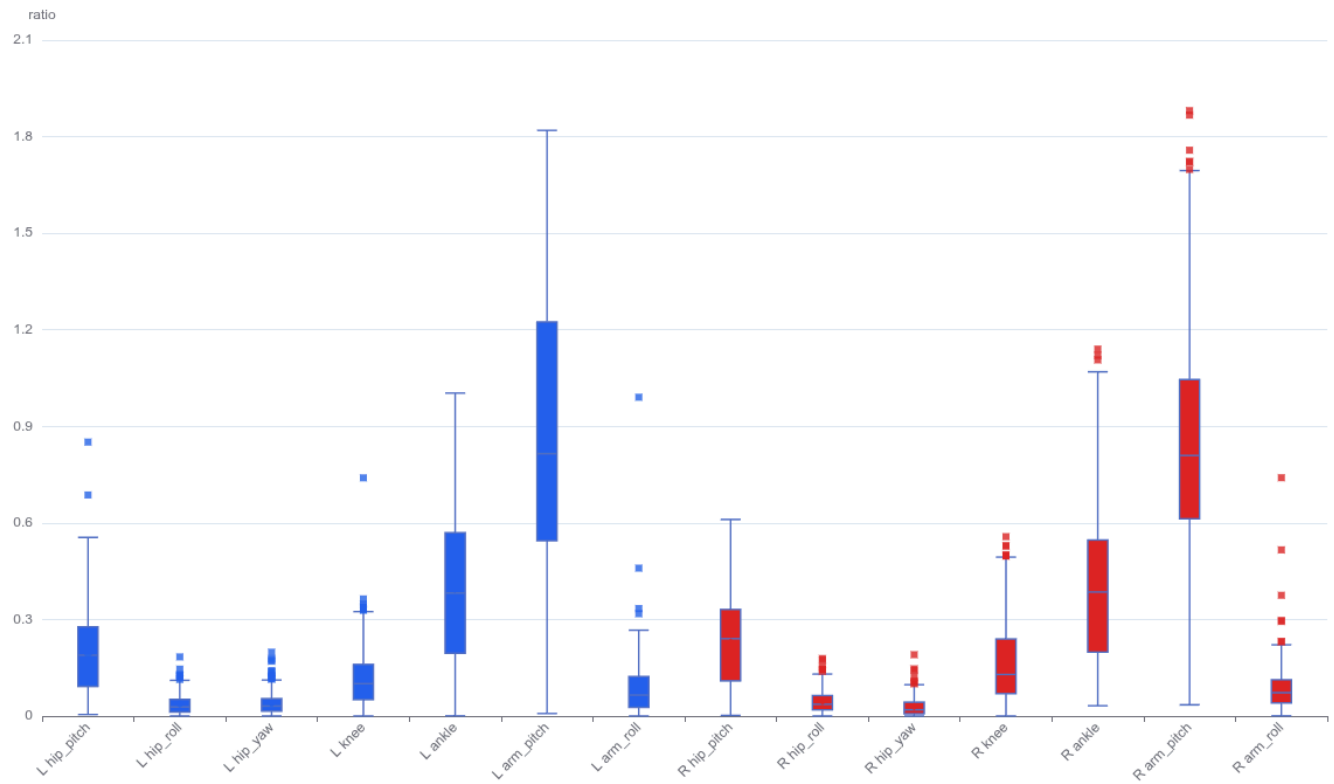

ratio
1
0.8
0.6
0.4
0.2
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

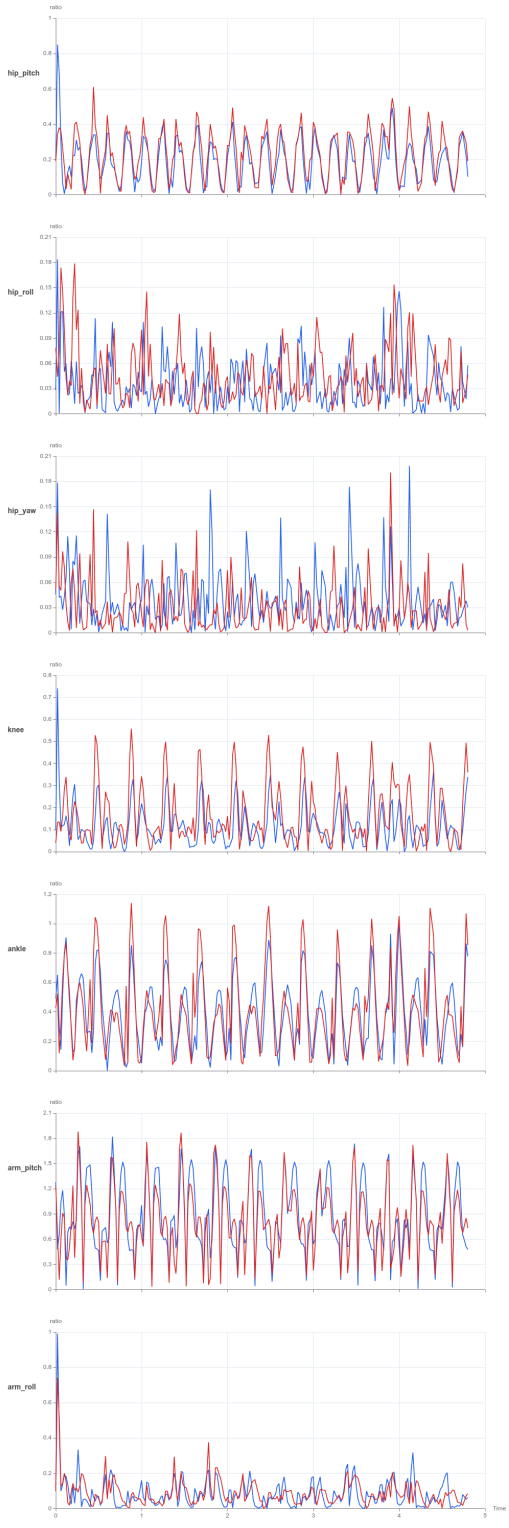

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
ratio
Time

关节速度占用



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_ratio/left_hip_pitch_joint	30	10	221	0.00483309	0.851245	0.191397	0.229522	0.851245	0.384323
joint_velocity_ratio/left_hip_roll_joint	10	15	221	0.000200099	0.183767	0.0373461	0.0490260	0.183767	0.103850
joint_velocity_ratio/left_hip_yaw_joint	10	15	221	0.0000449367	0.198480	0.0385594	0.0516318	0.198480	0.114804
joint_velocity_ratio/left_knee_joint	30	10	221	0.0000605958	0.740060	0.120984	0.154827	0.740060	0.306368
joint_velocity_ratio/left_ankle_joint	10	15	221	0.000541378	1.00357	0.394799	0.460937	1.00357	0.814398
joint_velocity_ratio/left_arm_pitch_joint	10	15	221	0.00803870	1.81956	0.880055	0.988715	1.81956	1.59201
joint_velocity_ratio/left_arm_roll_joint	10	15	221	0.000368839	0.990544	0.0870011	0.128024	0.990544	0.212526
joint_velocity_ratio/right_hip_pitch_joint	30	10	221	0.00209097	0.610851	0.228792	0.267124	0.610851	0.439879
joint_velocity_ratio/right_hip_roll_joint	10	15	221	0.000124853	0.178821	0.0452776	0.0568193	0.178821	0.118862
joint_velocity_ratio/right_hip_yaw_joint	10	15	221	0.0000524001	0.190692	0.0297679	0.0424534	0.190692	0.0919741
joint_velocity_ratio/right_knee_joint	30	10	221	0.0000961185	0.557404	0.171534	0.218418	0.557404	0.455680
joint_velocity_ratio/right_ankle_joint	10	15	221	0.0318561	1.14009	0.418167	0.504814	1.14009	0.991340
joint_velocity_ratio/right_arm_pitch_joint	10	15	221	0.0349508	1.88093	0.825481	0.921486	1.88093	1.58277
joint_velocity_ratio/right_arm_roll_joint	10	15	221	0.000567009	0.740443	0.0896697	0.119470	0.740443	0.198828

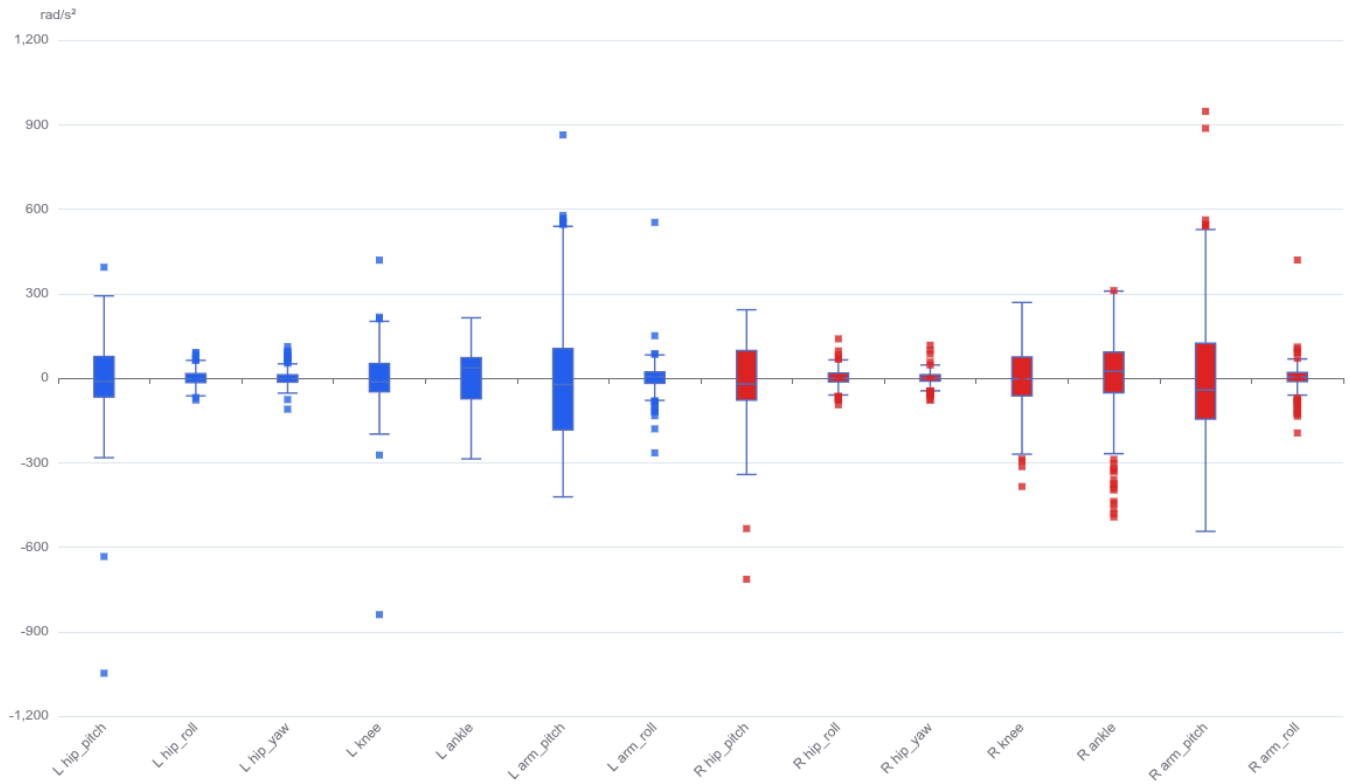

ratio
2.1
1.8
1.5
1.2
0.9
0.6
0.3
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

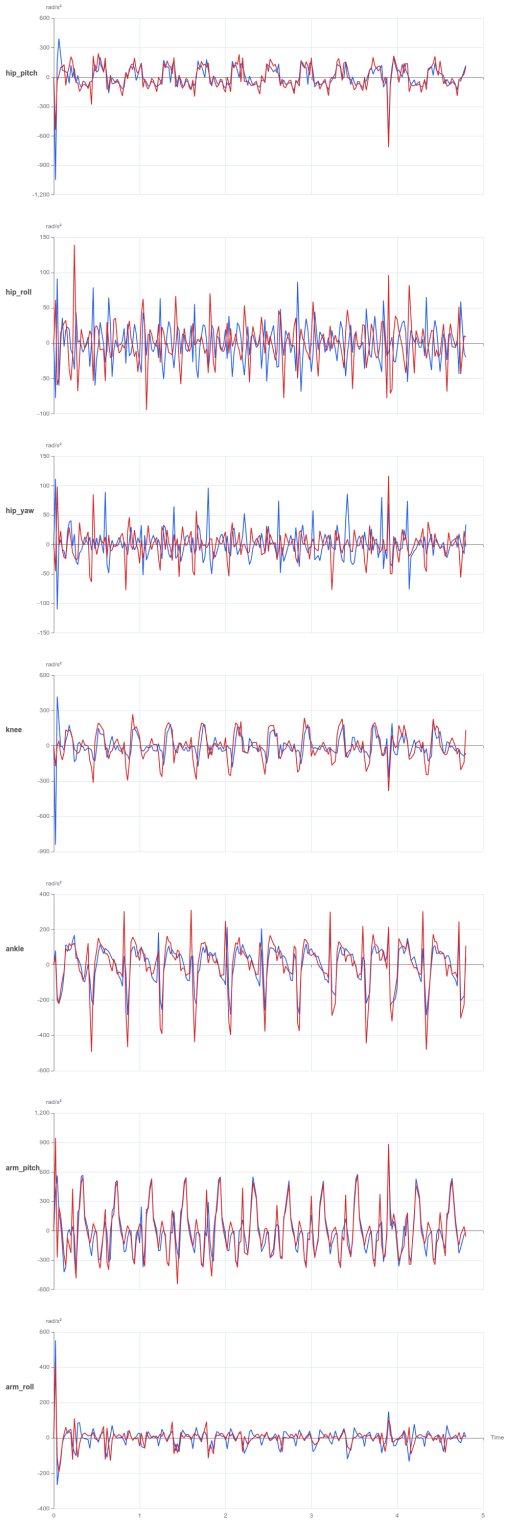

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
ratio
Time

关节加速度

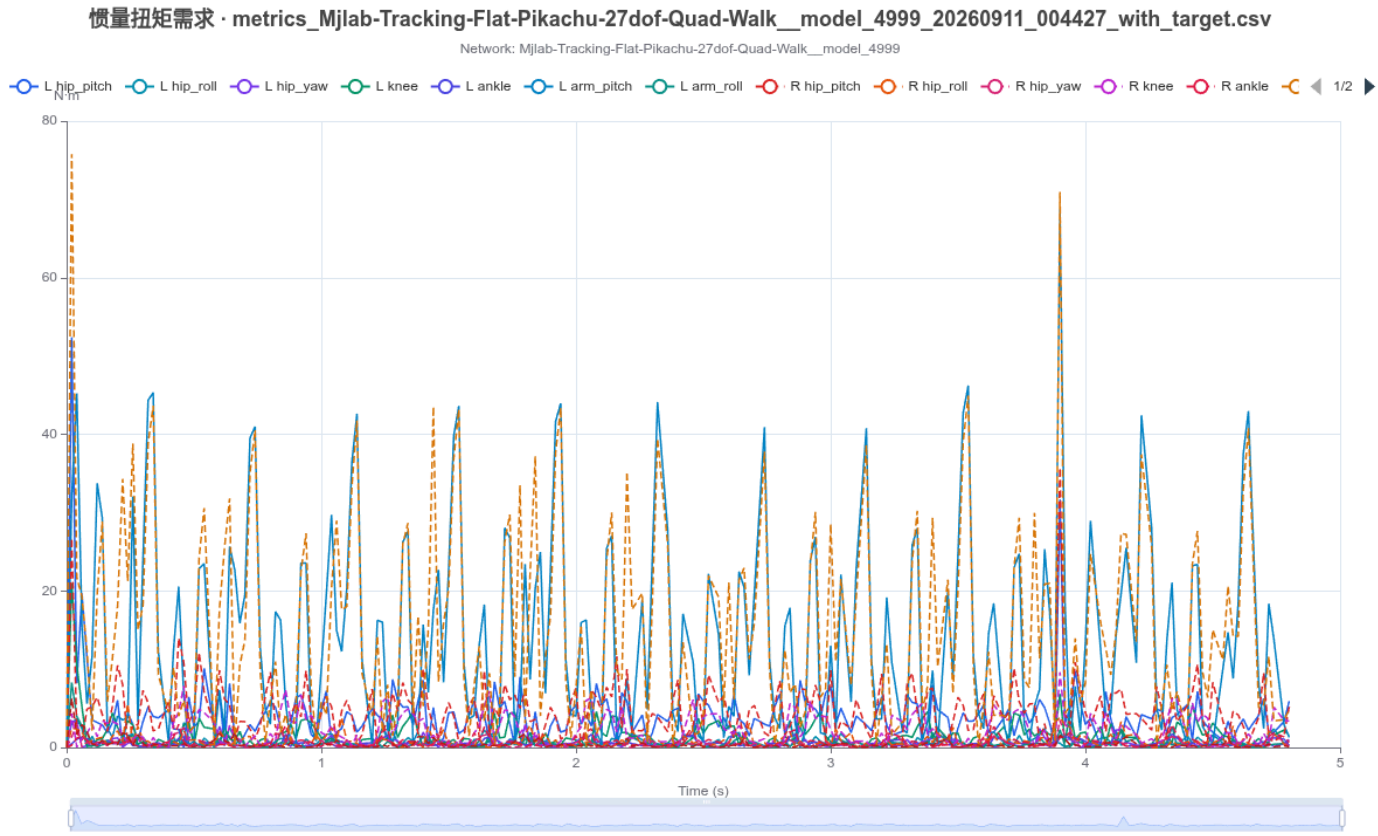


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_accel_rad_s2/left_hip_pitch_joint	30	10	221	-1047.99	393.759	0.358517	122.412	1047.99	164.355
joint_accel_rad_s2/left_hip_roll_joint	10	15	221	-77.7719	91.4686	-0.383420	27.4539	91.4686	58.9442
joint_accel_rad_s2/left_hip_yaw_joint	10	15	221	-110.312	111.897	0.548702	26.9006	111.897	57.9178
joint_accel_rad_s2/left_knee_joint	30	10	221	-839.835	418.786	-0.390811	102.666	839.835	176.008
joint_accel_rad_s2/left_ankle_joint	10	15	221	-286.756	214.008	0.0408490	106.992	286.756	218.364
joint_accel_rad_s2/left_arm_pitch_joint	10	15	221	-421.731	863.524	0.819465	241.659	863.524	530.362
joint_accel_rad_s2/left_arm_roll_joint	10	15	221	-265.487	553.010	-0.127619	58.2615	553.010	85.4517
joint_accel_rad_s2/right_hip_pitch_joint	30	10	221	-714.259	242.472	-0.203441	122.214	714.259	200.088
joint_accel_rad_s2/right_hip_roll_joint	10	15	221	-94.8208	139.354	0.459105	30.3042	139.354	65.5208
joint_accel_rad_s2/right_hip_yaw_joint	10	15	221	-77.6945	116.707	-0.297368	24.3337	116.707	52.3105
joint_accel_rad_s2/right_knee_joint	30	10	221	-384.962	268.404	-0.381147	119.052	384.962	237.826
joint_accel_rad_s2/right_ankle_joint	10	15	221	-493.185	310.718	-0.469159	145.355	493.185	329.881
joint_accel_rad_s2/right_arm_pitch_joint	10	15	221	-544.269	947.111	1.41036	251.822	947.111	488.010
joint_accel_rad_s2/right_arm_roll_joint	10	15	221	-194.794	419.272	0.219239	48.5667	419.272	92.5852

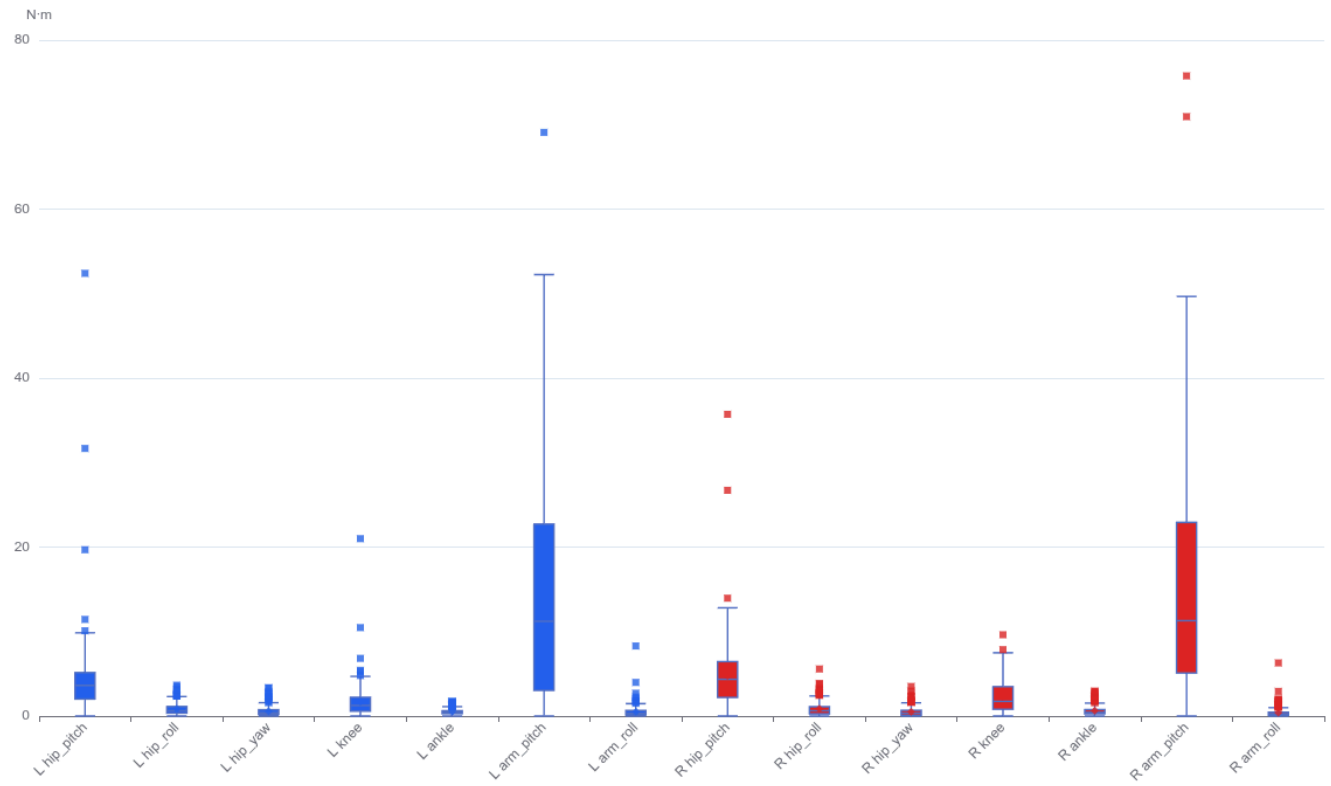

rad/s²
1,200
900
600
300
0
-300
-600
-900
-1,200
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

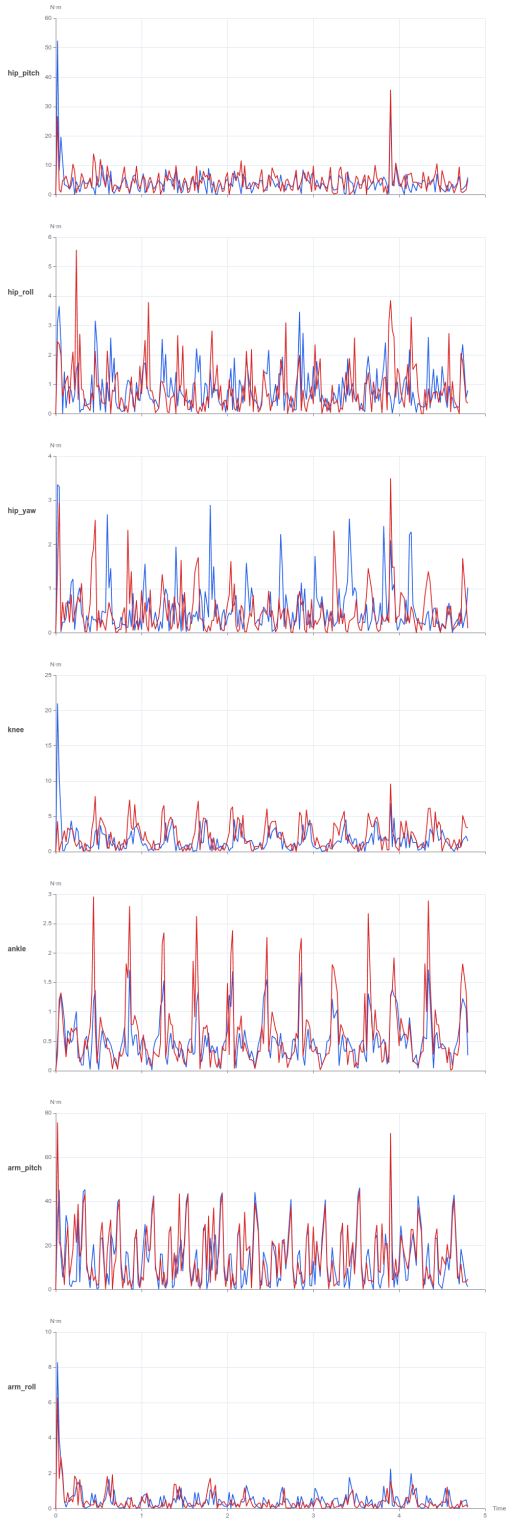

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
Time

惯量扭矩需求

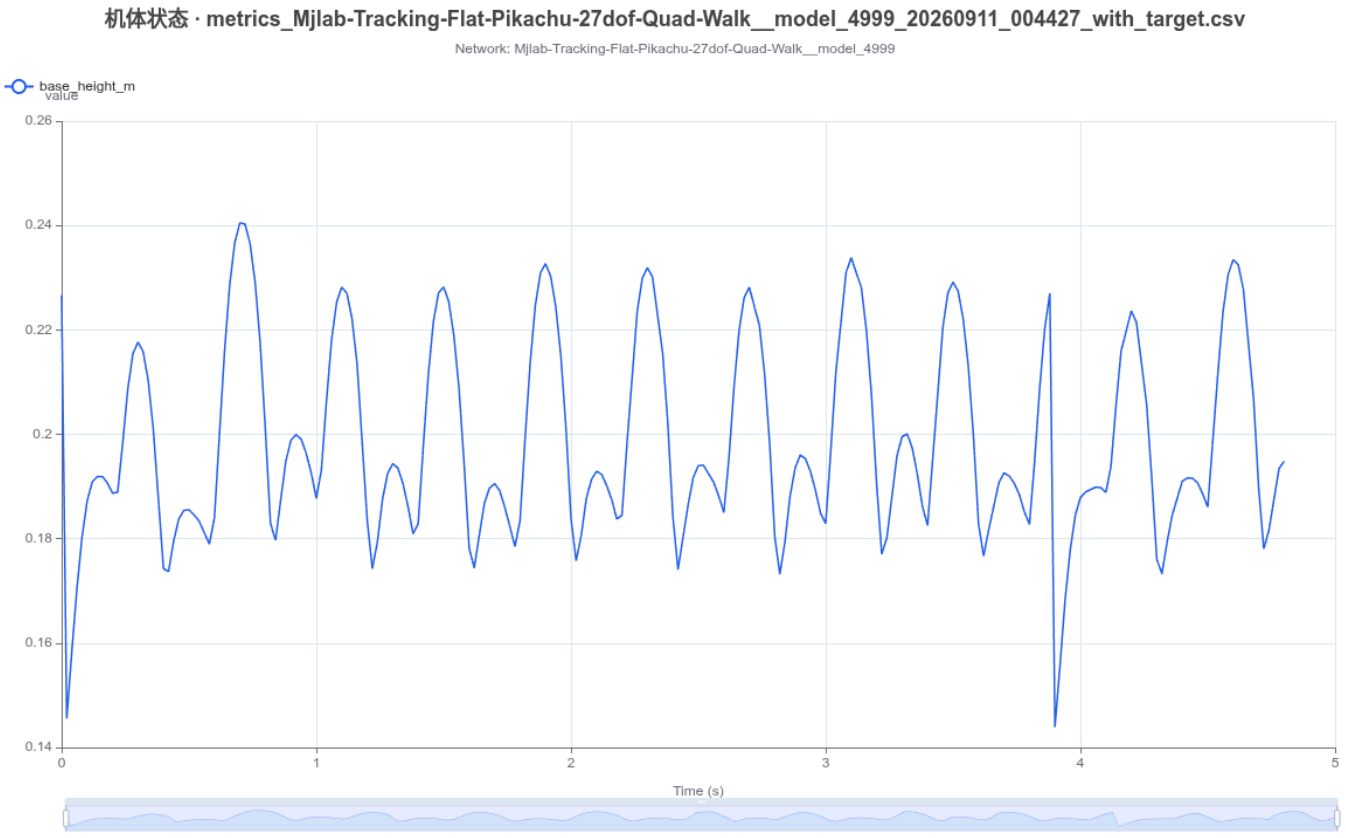


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_inertia_torque_nm/left_hip_pitch_joint	30	10	221	0.00000	52.3996	4.16106	6.12059	52.3996	8.21775
joint_inertia_torque_nm/left_hip_roll_joint	10	15	221	0.00000	3.65874	0.842568	1.09816	3.65874	2.35777
joint_inertia_torque_nm/left_hip_yaw_joint	10	15	221	0.00000	3.35692	0.559824	0.807017	3.35692	1.73753
joint_inertia_torque_nm/left_knee_joint	30	10	221	0.00000	20.9959	1.69651	2.56665	20.9959	4.40020
joint_inertia_torque_nm/left_ankle_joint	10	15	221	0.00000	1.72053	0.523170	0.641952	1.72053	1.31019
joint_inertia_torque_nm/left_arm_pitch_joint	10	15	221	0.00000	69.0819	14.3292	19.3327	69.0819	42.4289
joint_inertia_torque_nm/left_arm_roll_joint	10	15	221	0.00000	8.29515	0.492003	0.873922	8.29515	1.28177
joint_inertia_torque_nm/right_hip_pitch_joint	30	10	221	0.00000	35.7130	4.76258	6.11070	35.7130	10.0044
joint_inertia_torque_nm/right_hip_roll_joint	10	15	221	0.00000	5.57416	0.863459	1.21217	5.57416	2.62083
joint_inertia_torque_nm/right_hip_yaw_joint	10	15	221	0.00000	3.50122	0.495933	0.730012	3.50122	1.56931
joint_inertia_torque_nm/right_knee_joint	30	10	221	0.00000	9.62405	2.29292	2.97631	9.62405	5.94566
joint_inertia_torque_nm/right_ankle_joint	10	15	221	0.00000	2.95911	0.634221	0.872133	2.95911	1.97929
joint_inertia_torque_nm/right_arm_pitch_joint	10	15	221	0.00000	75.7689	15.3364	20.1457	75.7689	39.0408
joint_inertia_torque_nm/right_arm_roll_joint	10	15	221	0.00000	6.28908	0.412478	0.728500	6.28908	1.38878

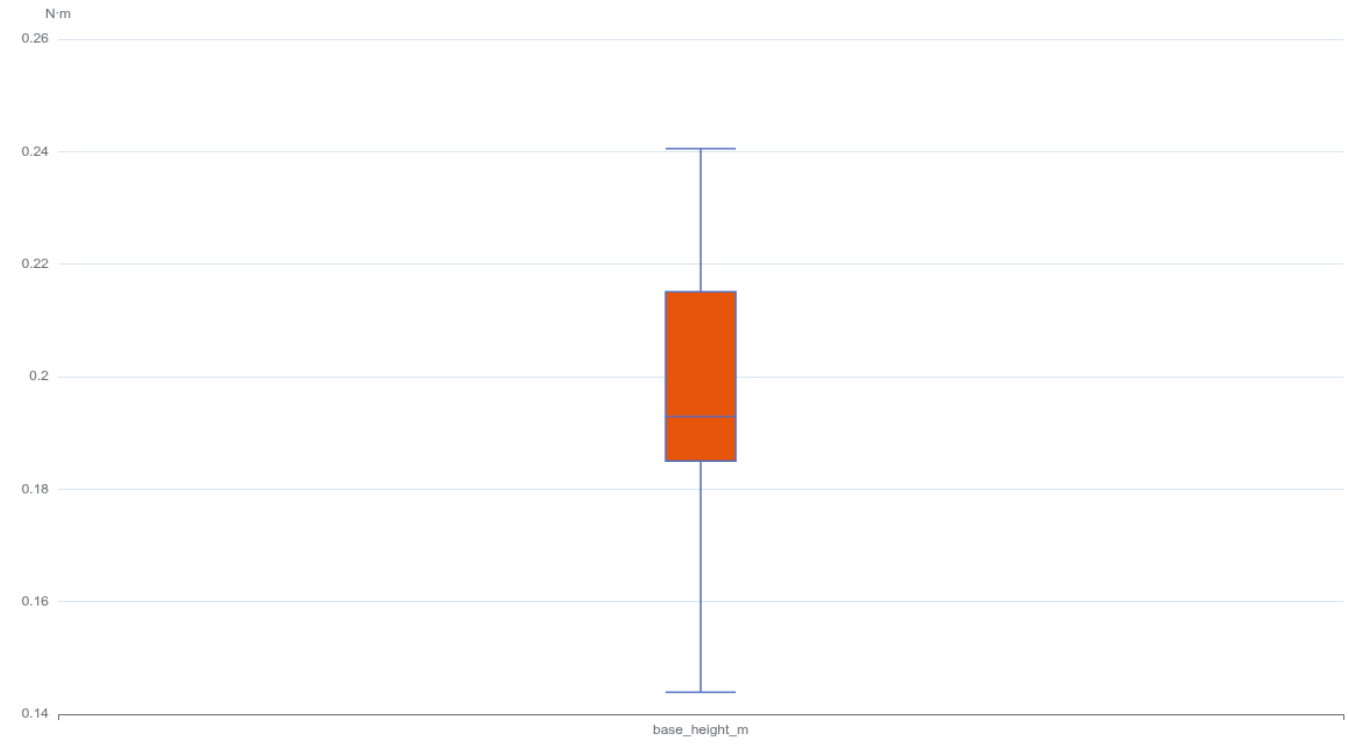

N·m
80
60
40
20
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll


hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
N·m
Time

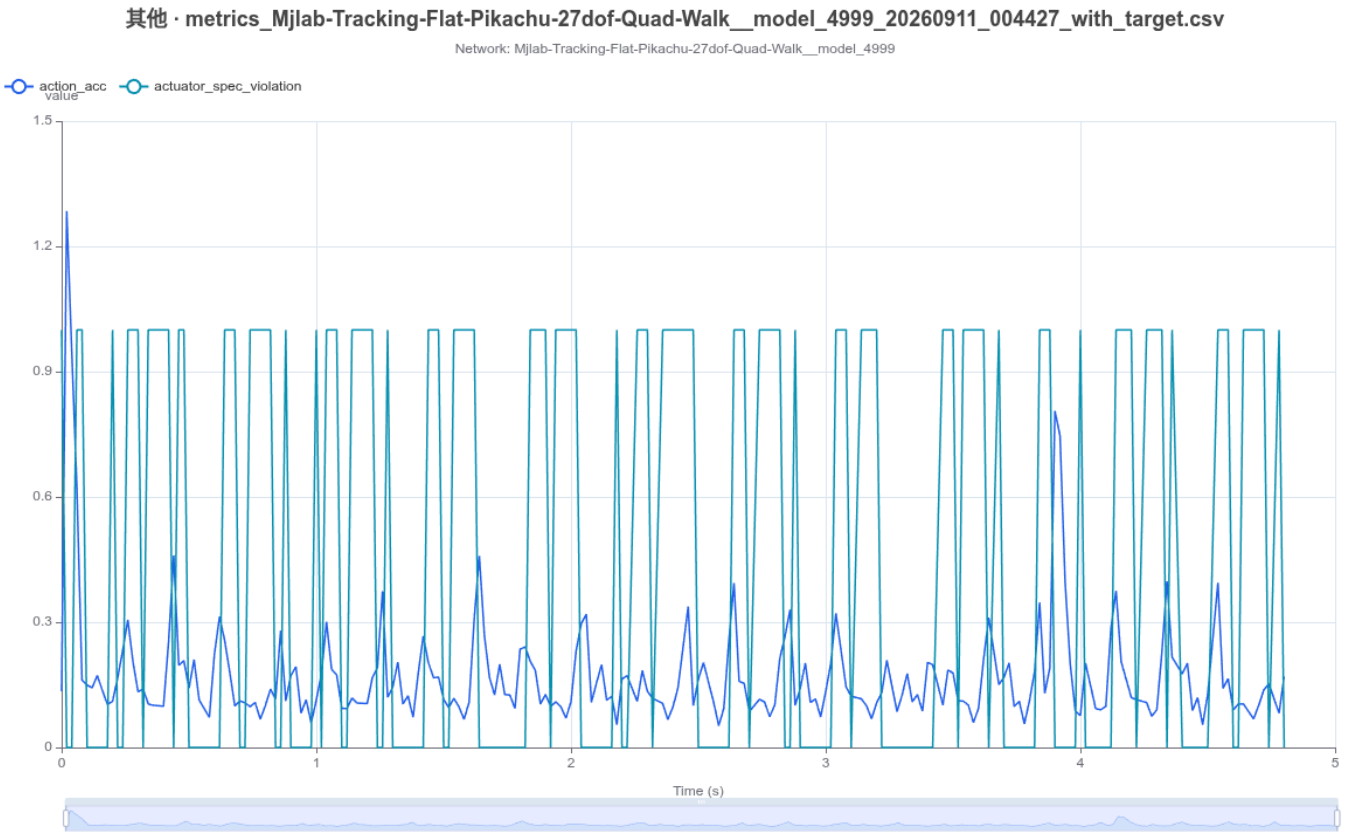
机体状态



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
base_height_m	–	–	221	0.143913	0.240531	0.198495	0.199397	0.240531	0.230590



其他



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
action_acc	–	–	221	0.0521121	1.28433	0.174063	0.222004	1.28433	0.375226
actuator_spec_violation	–	–	221	0.00000	1.00000	0.461538	0.679366	1.00000	1.00000

